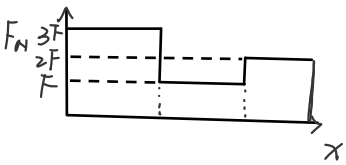
采用截面法：

① $F_1 \leftarrow 1 \rightarrow 2F$　　$F_1 = 2F$.

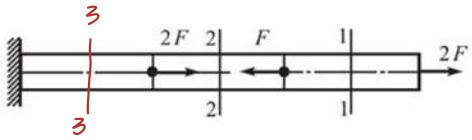
② $F_2 \leftarrow 2 \rightarrow 2F$　　$F_2 + F = 2F$　$\therefore F_2 = F$.

③ 对 3-3　$F_3 \leftarrow 2F \quad F \rightarrow 2F$

$F_3 + F = 2F + 2F$　　$F_3 = 3F$

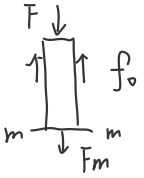


2-1　试求图2-2-1所示各杆1-1和2-2横截面上的轴力，并作轴力图。



2-2　一打入地基内的木桩如图2-2-3所示，沿杆轴单位长度的摩擦力为$f=kx^2$（k为常数），试作木桩的轴力图。

对m-m面：



$$\begin{cases} F + F_m = f_0 \\ f_0 = \int_x^l kx^2 dx \end{cases}$$

$\therefore F_m = \frac{1}{3}k(l^3 - x^3) - F$.

对整体分析.

$F = \int_0^l kx^2 dx = \frac{1}{3}kl^3$

$\therefore k = \frac{3F}{l^3}$

$\therefore F_m = -\frac{F}{l^3}x^3$

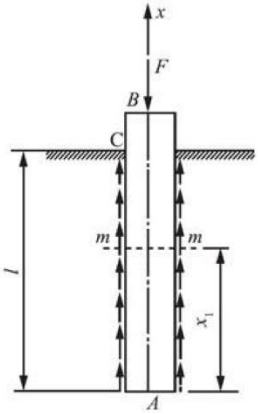
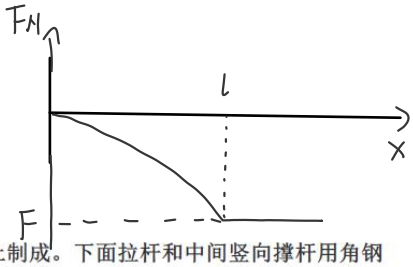


图2-2-3

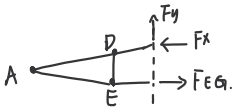


2-4　图2-2-5为一混合屋架结构的计算简图。屋架的上弦用钢筋混凝土制成。下面拉杆和中间竖向撑杆用角钢构成，其截面均为两个75mm×8mm的等边角钢。已知屋面承受集度为q＝20kN/m的竖直均布荷载。试求拉杆AE和EG横截面上的应力。

分布载荷合力：$F_0 = q \cdot l = 20 \times (2 \times 4.37 + 9) = 177.4\,kN$.

对整体，由对称性得 $F_A = \frac{1}{2}F_0$

对左半部分析：



对C取矩.

$F_{EG} \times (1+1.2) = F_A \times (4.37) + \frac{q}{2}$

$\therefore F_{EG} = 357.62\,kN$

$\therefore \sigma_{EG} = F_{EG}/(2A) = 357.62 \times 10^3 / (2 \times 11.5 \times 10^{-4})\,Pa = 155.5\,MPa$.

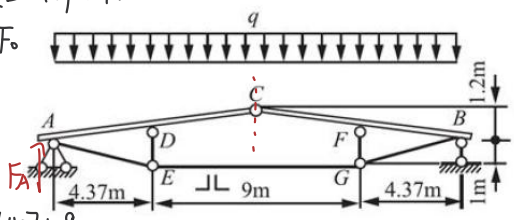
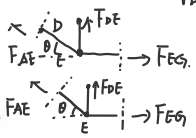


图2-2-5

对E点分析.　由平衡得



$F_{AE} \cdot \cos\theta = F_{EG}$

$\cos\theta = \frac{4.37}{\sqrt{4.37^2 + 1^2}}$

$\therefore F_{AE} = 366.86\,kN$

$\therefore \sigma_{AE} = F_{AE}/(2A) = 159.3\,MPa$

2-5　图2-2-7所示拉杆承受轴向拉力F＝10kN，杆的横截面面积A＝100mm²。如以α表示斜截面与横截面的夹角，试求：

（1）当α＝0°，30°，－60°时各斜面上的正应力和切应力，并用图表示其方向；

（2）拉杆的最大正应力和最大切应力及其作用的截面。

(1)　$\sigma_0 = \frac{F}{A} = \frac{10\,kN}{100\,mm^2} = 10^8\,Pa = 100\,MPa$.

① 当 $\alpha = 0$ 时.　$\sigma_\alpha = \sigma_0 \cos^2\alpha = \sigma_0 = 100\,MPa$

$\tau_\alpha = \frac{\sigma_0}{2}\sin 2\alpha = 0$.

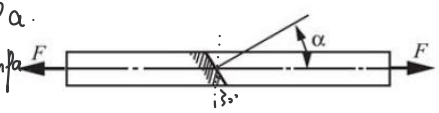


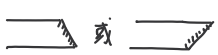
图2-2-7

② 当 $\alpha = 30°$ 时　$\sigma_\alpha = \sigma_0 \cos^2\alpha = \sigma_0 (\frac{\sqrt{3}}{2})^2 = 75\,MPa$

$\tau_\alpha = \frac{\sigma_0}{2}\sin 2\alpha = \frac{\sigma_0}{2}\sin 60° \approx 43.3\,MPa$.

③ 当 $\alpha = -60°$ 时.　$\sigma_\alpha = \sigma_0 \cos^2(-60°) = \frac{1}{4}\sigma_0 = 25\,MPa$.

$\tau_\alpha = \frac{\sigma_0}{2}\sin(-120°) = -\frac{\sigma_0}{2} \cdot \frac{\sqrt{3}}{2} = -43.3\,MPa$.

(2)　$\because \sigma_\alpha = \sigma_0 \cos^2\alpha \le \sigma_0$　当 $\alpha = 0$ 时. 取"="号.　　$\therefore$ 截面为 $\alpha = 0$ 时，即横截面

$\tau_\alpha = |\frac{\sigma_0}{2}\sin 2\alpha| \le \frac{\sigma_0}{2}$.　当 $\alpha = \pm\frac{\pi}{4}$ 时. 取"="号.　　$\therefore$ 截面为 $\alpha = \pm\frac{\pi}{4}$ 时.